Section 16

SUPPLY OF HARDWARE FITTINGS & ACCESSORIES FOR TRANSMISSION LINES SUITABLE FOR ACSR CONDUCTORS & HTGS WIRE

1.0 SCOPE :

This part of the specification provides for the manufacture, stage testing, inspection and testing before dispatch, supply and delivery of Hardware fittings and Accessories suitable for ACSR Conductors and HTGS wire specified herein for their satisfactory operation in 132kV and 220 kV Transmission lines.

2.0 STANDARDS :

2.1 The Hardware fittings and Accessories shall conform to the following Indian Standards, which shall mean latest revisions, amendments / changes adopted and published, unless otherwise specified hereinafter.

Sl. No.	Indian Standard	Title
1.	IS : 2486	Specification for Insulator fittings for overhead Power lines with nominal voltage greater than 1000 V. Part-I General Requirement and Tests Part-II Dimensional Requirements Part-III Locking Devices Part-IV Tests for locking devices
2.	IS: 2121	Specification of conductors and Earth wire Accessories for overhead Power lines. Part-I Armour Rods, Binding Wires and Tapes Part-II Mid Span Compression Joints and repair Sleeves. Part-III Accessories for earth wire.
3.	IS : 3138	Hexagonal Bolts & Nuts
4.	IS : 4218	Metric Screw Threads
5.	IS : 6639	Hexagonal Bolts for Steel
6.	IS : 4172	Dimensions for Heads of Bolts and Screws
7.	IS : 4206	Dimensions for nominal lengths & thread length for bolts, screws and studs.
8.	IS : 1570	Schedules for wrought steels for general engineering purposes. Part-I Steel specified by tensile and/ or yield properties.
9.	IS : 2004	Specification for Carbon Steel forging for general
10.	IS : 9708	Stock bridge Vibration Dampers.
11.	IS : 1573	Electroplated coatings of zinc on iron and steel.
12.	IS : 1367	Technical supply conditions for threaded fasteners
13.	IS : 2629	Technical supply conditions for threaded fasteners
14.	IS : 6745	Methods of determination of weight of Zinc-coating on Iron and steel.
15.	IS : 2633	Method for testing uniformity of coating of zinc coated articles.
16.	IS : 7814	Security Clip
17.	IS : 2326	Yoke Plate
18.	IS : 2016	Washers
19.	IEC : 372(2)	Test on locking device for ball and socket coupling
20.	IS : 2496	Galvanizing / Electroplating test
21.	IS : 209	Chemical analysis of zinc used for galvanizing
22.	IS:1363	Part-1 to 3 Hexagonal head Bolts, Screws and Nuts.

3.0 CLIMATIC CONDITIONS:

3.1 The Insulator Hardware shall be suitable for being installed directly in air between the insulator string and power conductor of 220 kV Double Circuit transmission lines with single Panther or single / Twin Moose conductor and 132 kV Double Circuit and Single

Circuit transmission lines with single Panther conductor and Earth wire. The conductor and earth wire accessories shall be suitable for being installed directly in air along with the power conductors and ground wire of double circuit and single circuit transmission lines. They shall be therefore suitable for satisfactory operation under the tropical climatic conditions listed in Section-I clause (3.2).

4.0 PRINCIPAL PARAMETERS :

4.1 DETAILS OF HARDWARE FITTINGS :

4.1.1 The Hardware fittings shall be as per the specification drawings attached. Conductor hardware fittings shall be suitable for single/double suspension insulator strings and single/double tension insulator strings.

4.1.2 Earth wire hardware fittings shall be suitable for the earth wire whose technical parameters are furnished in Annexure-A.

4.1.3 Each hardware fittings shall be supplied complete in all respects and shall include all components indicated in the specification drawings.

4.2 DETAILS OF ACCESSORIES:

i) The Power Conductor accessories shall be suitable for the ACSR conductors whose technical parameters are furnished in Annexure-A.

ii) The ground wire accessories fittings shall be suitable for the ground wire whose technical parameters are furnished in Annexure-A.

4.3 INSULATOR STRING CHARACTERISTICS :

4.3.1 The characteristics of the complete string shall be as follows:

Sl No	Details	220 kV				132 kV			
		Suspension		Tension		Suspension		Tension	
		Single	Double	Single	Double	Single	Double	Single	Double
1	Lightning impulse withstand voltage (dry) kV peak	1050	1050	1050	1050	650	650	650	650
2	Power frequency withstand voltage (wet) kV rms	460	460	460	460	280	280	280	280
3	Mechanical failing load - kgf.	7000	14000	11500	23000	7000	14000	11500	23000
4	No deformation load – kgf. (67% of MFL)	4690	9380	7705	15410	4690	9380	7705	15410

4.4 SPECIFICATION DRAWINGS:

A list of specification drawings in respect of the Hardware fittings and Accessories indicated above is given in Volume-III of the bid document as mentioned at Section 1(clause 7) of this volume. These specification drawings are enclosed for information and guidance of the Bidder. The Bidder shall furnish Hardware string drawings along with component drawings in accordance with specification drawings. The Bidder shall furnish fabrication drawings with all additional dimensions along with the critical dimensions specified on the specification drawings. Material specifications details of applicable standards and heat treatment details shall also be furnished. The details of manufacturing process also shall be furnished along with the component drawings. The details of trade mark to be provided on the components shall also be furnished.

The Bidder should not deviate from the dimensions and weights given in the General Technical Requirements. The bids having no drawings or incomplete drawings or

drawings with lesser dimensions are liable for rejection. The materials to be supplied must be as per the final drawings approved by this office after placing the Order. The successful bidder has to furnish the approved final drawings in CD.

5.0 GENERAL TECHNICAL REQUIREMENTS :

The suspension hardware suitable for Moose / Twin Moose and Panther ACSR Conductor to be supplied shall consist of the following.

Sl. No.	Item		Material
I.	**FOR SINGLE SUSPENSION HARDWARE**		
1.	Ball Hook	-	Drop forged steel to class 4 as per IS-2004
2.	Socket eye	-	Drop forged steel to Class 4 as per IS-2004 (Malleable cast iron not acceptable)
3.	Arcing horns	-	Tubular steel (Tower side & Lone side)
	Suspension Clamp (Armour Grip type)		
	a) Body (Support Housing)	-	Heat-treated gravity Die casting high strength corrosion resistant Aluminium Alloy.
	b) Clamping Components Elastomer insert with forged Aluminum Alloy Reinforcement	-	Ozone resistant neoprene rubber having shore hardness 65 to 80/Forged Aluminum Alloy.
	c) Supporting Strap Alloy Aluminum as per IS-733. Grade-65032		-Heat-treated high strength extruded/ forged
	d) Bolts and Nuts	-	As per IS-6639 galvanised to IS-1367 (GKW or equivalent)
	e) Flat Washer	-	M 16 size as per IS-2016
	f) Spring Washer	-	Flat section as per IS-1573 Electro Galvanised to service conditions.
	g) Security clip	-	Security clip made of phosphor bronze as per IS-8714Grade-1. Clips of Brass, chemical Bronze etc. not acceptable.
5.	Retaining rods (P.A.Rods)	-	Heat-treated Aluminum Alloy.

5.1 DIMENSIONS OF INSULATOR STRING ALONG WITH HARDWARE FITTINGS:

5.1.1 The various dimensions of insulator strings shall be as per the specification drawings.

5.1.2 Fabrication drawings shall be prepared by the manufacturer based on the specification drawings.

5.1.3 The dimensions which are not provided on the specification drawings shall be suitably selected by the manufacturer while preparing the fabrication drawings.

5.1.4 The fabrication drawings shall be submitted to the purchaser for scrutiny and approval.

5.1.5 Dimensional tolerance as indicated on the specification drawings only shall be allowed.

5.2 CORONA AND RI PERFORMANCE :

Sharp edges and scratches shall be avoided on all the hardware fittings. All surfaces must be clean, smooth, without cuts and abrasions or projections. The Bidder must give

suitable assurance about the satisfactory corona and radio interference performance of the materials offered by him.

5.3 BALL AND SOCKET FITTINGS :

The dimensions of the ball and socket shall be in accordance with standard dimensions stated in IS:2486 (Part-II), as mentioned below.

Details	Suspension	Tension
Panther	16 'B'mm	20mm
Moose	16 B'mm	20mm
Twin Moose	20 mm	20mm

5.3.1 Ball Fittings :

Ball fittings shall be made of class IV steel as per IS:2004 or steel of equivalent grade forged in one piece. They shall be normalised to achieve the minimum breaking strength specified on the respective drawings. Before galvanisation of ball fittings, all die flashing on the shank and on the bearing surface of the ball shall be carefully removed without reducing the dimensions below the requirements.

5.3.2 Socket Fittings :

Socket fittings shall be made of class IV steel as per IS:2004 or steel of equivalent grade and shall be forged in one piece. They shall be normalised to achieve the minimum breaking strength specified on the respective drawings.

5.3.3 Dimensions and Tolerances :

5.3.3.1 The dimensions and tolerances of pin balls and socket ends shall conform to IS:2486 Part-II/IEC 120 and shall be checked by the gauge therein after galvanising.

5.3.3.2 The pin balls shall be checked with the applicable 'GO' gauges in at least two directions, one of which shall be across the line of die flashing, and the other 90 Deg to this line. "NO GO" gauges shall not pass in any direction.

5.3.3.3 The bearing surfaces of balls and machined sockets' before galvanising, shall not have surface roughness more than 250 micro inches.

5.3.3.4 The bearing surface of socket ends shall be uniform about the entire circumference without depressions or high spots. The internal contour of the socket ends shall be concentric with the axis of fittings. The axis of the bearing surface of socket ends shall be coaxial with the axis of fittings with no appreciable tilting.

5.3.4 SECURITY CLIPS (LOCKING DEVICES):

5.3.4.1 Socket fitting shall be provided with security clip in accordance with IS:2486 (Parts III & IV) to provide positive locking against unintentional disengagement of socket from the ball of the insulator.

5.3.4.2 The hole for the security clip shall be on the side of the socket opposite to the socket opening. The hole for the clip shall be countersunk. The clip shall be of such design that the same may be engaged by a hotline clip puller to provide for disengagement under energised conditions.

5.3.4.3 The security clip shall be made of phosphor bronze conforming to IS:7814 or of brass conforming to IS 410 with a minimum hardness of 160 HV for Split Pins and 150 HV for W-Clips.

5.4(a) YOKE PLATE/LINK PLATE :

5.4.1 The yoke plates for Suspension and Tension Hard Ware suitable for Twin Moose shall be made of mild steel plate as per IS:226 or equivalent standards. Shearing/cutting of the plates shall be clean without drawn or ragged edges. If the plates are flame cut, mechanical guides shall be used. It shall be ensured that the grain flow of the yoke plate shall be in the direction of the tensile load.

5.4.2 Holes shall be cylindrical, clean cut and perpendicular to the plane of the material. The periphery of the holes shall be free from burrs.

5.4.3 All the corners and edges should be rounded off with a radius of at least 3 mm.

5.4(b) SAG ADJUSTMENT DEVICE (For Twin Moose Towers)

5.4.4 The sag adjustment device for 220KV lines shall be of high quality mild steel or high tensile steel and hot dip galvanised.

5.4.5 The sag adjustment device to be provided with the double tension fitting shall be of the three-plate type. The sag adjustment device shall be provided with a safety locking arrangement. The device shall be of such design that the adjustment is done with ease, speed and safety. The ultimate tensile strength shall be 161.78 KN (16,500 Kg.).

5.4.6 The maximum length of sag adjustment device from the connecting part of the rest of the hardware fittings shall be 520 mm. The details of maximum and minimum adjustment shall be clearly indicated in the drawing submitted with tender. The adjustment of 150 mm shall be possible with the sag adjustment device. The step of adjustment shall not be more than 15 mm.

5.5 ARCING HORNS :

5.5.1 Arcing Horns of fixed type and adjustable type should be with double bolt fixing arrangement as shown in the drawings enclosed.

5.6 SUSPENSION ASSEMBLY :

5.6.1 The suspension assembly shall be suitable for **Single & Twin** ACSR 'MOOSE' Conductor for 220 kV lines and Single ACSR 'PANTHER' Conductor for 132 kV lines.

5.6.2 For ACSR Conductor the suspension assembly shall include armour grip suspension clamp.

5.6.3 The suspension clamp along with standard preformed armour rods set shall be designed to have maximum mobility in any direction and minimum moment of inertia so as to have minimum stress on the conductor in the case of oscillation of the same.

5.6.4 The suspension clamp along with standard preformed armour rods set shall have a slip strength as per relevant standards.

5.6.5 The suspension assembly shall be smooth without any cuts, grooves, abrasions, projections ridges or excrescence which might damage the conductor.

5.6.6 Armour Grip Suspension Clamp :

5.6.6.1 The armour grip suspension clamp shall comprise of retaining strap, support housing, elastomer inserts with aluminium reinforcements and AGS preformed rod set as indicated on the specification drawings.

5.6.6.2 The armour grip suspension clamp shall be so manufactured that it shall minimise the static and dynamic stress developed in the conductor under various loading conditions as well as during wind induced conductor vibrations. It shall allow slipping of the conductor under unbalanced conductor tension in adjacent spans and broken wire conditions. It shall also withstand power arcs and have required level of Corona/RIV performance.

5.6.6.3 Elastomer cushion shall be resistant to the effects of temperature from -40 deg.C to +75 deg.C, ozone, ultraviolet radiations and other atmospheric contaminants likely to be encountered in service. The physical properties of the elastomer shall be of approved standard. It shall be electrically shielded by a cage of AGS preformed rod set. The elastomer cushion shall be so designed that the curvature of the AGS rod shall follow the contour of the neoprene cushion and shall not leave any gap between the two when installed.

5.6.6.4 The support housing of AGS clamp shall be forged out of aluminium alloy of type 6061 or equivalent. It shall be duly heat treated to achieve the specified tensile properties and to maintain consistent material characteristics during service.

5.6.6.5 The aluminium reinforcement of neoprene cushion shall be extruded and formed out of aluminium alloy of type 6061 or equivalent. It shall be duly heat treated for the reasons mentioned under clause 5.6.6.4.

5.6.6.6 The retaining strap shall be extruded and then formed out of aluminium alloy of type 6061 or equivalent. It shall be duly heat treated for the reasons mentioned under clause 5.6.6.4.

5.6.6.7 The elastomer cushion shall be moulded on the aluminium reinforcement.

5.6.7 AGS Preformed Armour Rods :

The AGS Preformed Armour rod set suitable for ACSR Conductor shall be used to minimise the stress developed in the conductor due to different static and dynamic loads because of vibration due to wind, slipping of conductor from the suspension clamp as a result of unbalanced conductor tension in adjacent span and broken wire conditions. It shall also withstand power arcs, chafing and abrasion from suspension clamp and localised effect due to resistance losses of the conductor.

5.6.7.1 The preformed armour rod set shall have right hand lay and the inside diameter of the helics shall be less than the outside diameter of the conductor to grip the same tightly. The surface of the armour rod when fitted on the conductor shall be smooth and free from projections, cuts and abrasions etc.,

5.6.7.3 The pitch length of the rods shall be determined by the Bidder but shall be less than that of the outer layer of ACSR conductor and the same shall be accurately controlled to maintain uniformity and consistently reproducible characteristic wholly independent of the skill of linemen.

5.6.7.4 The length and diameter of each rod shall be as per the values given below. The tolerance in length of the rods in completed set should be within 13 mm between the longest and shortest rod. The ends of armour rod shall be parrot billed.

5.6.7.4.1 PREFORMED ARMOUR RODS :

Description	**MOOSE**	**PANTHER**
i) Breaking strength	35 kg/mm2	35 kg/mm2
ii) No. of rods per set	12	11
iii) Diameter of rod with tolerances (mm.)	9.27 +0.0 or –0.1	6.35 +0.0 or –0.1
iv) Length of rod with tolerances (mm.)	2235 +0 or -51	1520 +0 or - 38

5.6.7.5 The number of armour rods in each set shall be twelve for 'MOOSE' conductor and eleven for 'PANTHER' conductor. Each rod shall be marked in the middle with paint for easy application on the line.

5.6.7.6 The armour rod shall not lose their resilience even after five applications.

5.6.7.7 The conductivity of each rod of the set shall not be less than 40% of the conductivity of the International-Annealed Copper Standard (IACS).

5.7 DEAD END ASSEMBLY :

5.7.1 The dead end assembly shall be suitable for ACSR single 'MOOSE'/Twin Moose conductor, for 220 kV lines and ACSR Single 'PANTHER' conductor for 132 kV lines.

5.7.2 The dead end assembly shall be of compression type with provision for compressing jumper terminal at one end. The angle of jumper terminal to be mounted should be 30 Deg. with respect to the vertical line. The area of bearing surface on all the connections shall be sufficient to ensure positive electrical and mechanical contact and avoid local

heating due to copper losses. The resistance of the clamp when compressed on the conductor shall not be more than 75% of the resistance of equivalent length of conductor.

5.7.3 Die compression length shall be clearly marked on each dead end assembly designed for continuous die compressions and shall bear the words **"COMPRESS FIRST"** suitably inscribed near the point on each assembly where the compression begins. It shall bear identification marks **"COMPRESSION ZONE"** and **'NON-COMPRESSION ZONE'** distinctly with arrow marks showing the direction of compressions and knurling marks showing the end of the zones. Tapered aluminium filler plugs shall also be provided at the line of demarcation between compression and non-compression zone. The letters, number and other markings on the finished clamp shall be distinct and legible. The dimensions and dimensional tolerances of the cross section of aluminium and steel dead ends shall be as per the specification drawings.

5.7.4 The assembly shall not permit slipping of, damage to or failure of the complete conductor or any part thereof at a load less than 95% of the minimum UTS of the conductor.

5.7.5 The outer sleeve shall be made out of EC grade aluminium 99.5% pure. The steel sleeve shall be made of mild steel rod.

5.8.0 HARD WARE FOR GROUND WIRE

5.8.1 SUSPENSION HARDWARE FOR GROUND WIRE:

i) Suspension hardware shall be used for suspending ground wire at all tangent/suspension towers and shall be suitable for supporting the GSS ground wire of size 7/3.15 mm for 220 kV and 132 kV lines.

ii) The suspension clamps shall conform to the latest version of IS:2486 and shall have adequate area of support for the ground wire. The groove of the clamp shall be smooth, finished in a uniform circular or oval shape and shall slope downwards in a smooth curve to avoid edge support and hence to reduce the intensity of bending moment on the ground wire.

iii) There shall be no sharp points in the clamps coming in contact with ground wire. There shall not be any displacement in the configuration of the ground wire strands nor shall these be unduly stressed in final assembly.

iv) The clamping piece and the clamp body shall be clamped by at least two "U" bolts of size not less than 10mm with 3mm thick lock washers on each of its limbs. Suspension clamp shall be provided with inverted type of "U" bolts. One limb of the "U" bolt shall be long enough to accommodate the lug of the flexible steel bond.

v) The suspension clamp shall be the "Trunion" type or "envelope" type.

vi) The complete assembly of the suspension clamp shall be guaranteed for slip strength of not less than 15 kN and not more than 17 kN. The ultimate tensile strength of the assembly including components shall not be less than 70 kN.

5.8.2 TENSION HARDWARE FOR GROUND WIRE:

i) Tension hardware shall be used at all tension towers for anchoring the 7/3.15mm galvanised steel ground wire for 220 kV and 132 kV lines. The hardware assembly shall comprise of compression type tension clamp and two "D" shackles. The tension clamp shall be attached to the horizontal strain plate of the tower body by means of a "D" shackle.

ii) These clamps shall give adequate area of support without any slip to the ground wire under normal working tension and vibration conditions.

iii) The complete tension hardware assembly shall be so designed as to avoid undue bending in any part of the clamp and shall not produce any hindrance to the movements of the clamp in horizontal and vertical direction.

iv) The slip strength of tension clamp assembly shall not be less than 95% of the ultimate strength of the ground wire. The ultimate strength of the clamp and individual components shall not be less than that of ground wire.

5.9 ACCESSORIES FOR CONDUCTOR:

5.9.1 MID SPAN COMPRESSION JOINTS FOR POWER CONDUCTOR:

i) This shall be suitable for joining the two ends of MOOSE ACSR conductor for 220 kV lines and PANTHER ACSR conductor for 132 kV lines. The joint shall have conductivity more than the conductivity of an equivalent length of the conductor. The joint shall not permit slipping of, damage to, or failure of the complete conductor or any part thereof at a load of not less than 95% of the ultimate tensile strength of the conductor. The electrical resistance of the joint, after installation, shall not exceed 75% of the measured resistance of the equivalent length of the conductor.

ii) The components of the joint for ACSR conductor shall consist of steel and aluminium sleeves for joint compression of the steel core conductor and aluminium conductor respectively. The steel sleeve shall not crack or fail during compression. The Brinnel Hardness number of the steel sleeves shall not exceed 200. The steel sleeve shall be hot dip galvanised. The aluminium sleeve shall be manufactured and extruded out of EC grade aluminium with a purity of not less than 99.5%. Tapered aluminium filler plugs shall be provided at the line of demarcation between compression and non-compression zones.

iii) The dimensions and dimensional tolerances of the mid span compression joint shall be as per the values given below.

Description	MOOSE	PANTHER
Diameter of Sleeves (mm)		
a) Before Compression		
(Inner Diameter)		
i) Aluminum	34 + or – 0.5	23 + or – 0.5
ii) Steel	11 + or – 0.5	9.45 + or- 0.5
(Outer Diameter)		
i) Aluminum	54 + or – 1	38 + or – 0.5
ii) Steel	24 + or – 0.5	18 + or – 0.5
b) After Compression		
(Corner to Corner)		
i) Aluminum	53 + or – 0.5	37 + or – 0.5
ii) Steel	23.4 + or – 0.5	17.4 + or – 0.5
(face to Surface)		
i) Aluminum	46 + or – 0.5	32 + or - 0.5
ii) Steel	20.2 + or – 0.5	15.1 + or – 0.5
Length of Sleeve (mm)		
a) Before Compression		
i) Aluminum	735 + or – 5	610 + or – 5
ii) Steel	305 + or – 5	203 + or – 5
b) After Compression		
i) Aluminum	785 + or – 5	650 + or – 5
ii) Steel	330 + or - 5	225 + or - 5

5.9.2 REPAIR SLEEVES FOR POWER CONDUCTOR:

i) Repair sleeve shall be used for repairing the conductor when a few strands of the aluminium conductor in the outermost layer are damaged with scratches, kinks, abrasions, nicks or cuts. Sleeve shall be of the compression type. The sleeve shall be manufactured and extruded out of EC grade aluminium having a purity of 99.5%. The sleeve shall be in two halves with a seat provision for sliding of the keeper piece. The edges of the seat as well as of the keeper piece shall be so rounded that the conductor strands are not damaged during installation. The outer body of the sleeve shall be smooth, even and with rounded off edges. The compressed conductor with the repair sleeve shall not permit damage or failure of the conductor at a load of not less than 95% of the ultimate tensile strength of the conductor. The electric resistance of the repaired portion of the conductor shall not exceed 75% of the measured resistance of an equivalent length of the conductor.

ii) The dimensions and dimensional tolerance of the repair sleeves shall be as per values given below.

Description	MOOSE	PANTHER
Diameter of Sleeves (mm)		
a) Before Compression		
Outer Diameter	54 + or – 1	38 + or – 1
b) After Compression		
i) Corner to Corner	53 + or – 0.5	37 + or – 0.5
ii) Surface to Surface	46 + or – 0.5	32 + or – 0.5
a) Before Compression	305 + or – 5	241 + or – 5
b) After Compression	330 + or - 5	271 + or - 5

5.9.3 VIBRATION DAMPERS FOR POWER CONDUCTOR:

i) The vibration dampers shall be of either 4R type or stock-bridge type for being used at all suspension and tension points at each and every span to damp out the vibrations of the conductors to the level specified hereinafter.

ii) The clamp of the vibration damper shall be made of aluminium alloy. It shall be capable of supporting the damper during installation and prevent damage or chafing of the conductor during erection or continued operation. The clamp shall have sufficient grip to maintain the damper in position on the conductor without damaging the strands or causing premature fatigue to the conductor under the clamp. The groove of the clamp body and clamp cap shall be smooth, free of projections, grit or other materials, which could cause damage to the conductor when the clamp is installed. Clamping bolts shall be provided with self locking nuts and designed to prevent corrosion of the threads or loosening during service.

iii) The messenger cable of the damper shall be made of high strength steel with a minimum strength of 136 kg/sq.mm. It shall be post formed in order to prevent subsequent droop of weight in service and to maintain consistent flexural stiffness of the cable while in service. The messenger cable shall be suitably and effectively sealed to prevent corrosion.

iv) The damper mass shall be made of hot dip galvanised mild steel/cast iron or a permanent mould cast zinc alloy. All castings shall be free from defects such as cracks, shrinkages, inclusions and blowholes etc. The inside and outside surfaces of the damper masses shall be smooth.

v) The damper assembly shall be electrically conductive to reduce radio interference.

vi) The vibration damper shall be capable of being installed and removed from energised line by means of hot line technique. In addition, the clamp shall be capable of being removed and reinstalled on the conductor at the design torque without shearing or damaging of bolts, nuts or cap screws.

vii) The vibration analysis report of the system, with and without damper, as detailed under **Annexure-B** shall be submitted by the Bidder along with the Bid. The technical particulars for vibration analysis and damping design of the system are as follows:

Sl. No.	Description	Technical Particulars of	
		220 kV Lines	132 kV Lines
1.	**Configuration**	Single ACSR Moose Conductor per phase in vertical formation	Single ACSR Panther Conductor per phase in vertical formation

		(or) Double Circuit Twin ACSR Moose Conductor bundle per phase in Horizontal formation at 400 mm apart and all three phases of each circuit in vertical formation on each side of tower.	
2.	**Span length in metres**		
i)	Ruling design span	350	300
ii)	Maximum span	400	400
iii)	Minimum span	100	125
3.	**Nature of terrain**	Partly hilly, partly plain with cultivated fields	Partly hilly, partly plain with cultivated fields
4.	**Armour rods**	Standard preformed armour rods/ AGS clamp	Standard preformed armour rods/ AGS clamp
5.	**Maximum wind load (m/sec)**	50 - Acting on full projected area of single conductor	50 - Acting on full projected area of single conductor
6.	**Max. permissible dynamic strains (micro strains)**	150	150

Technical Requirements for 4R Vibration Dampers for ACSR Conductor

Description	MOOSE	PANTHER
a) Total weight of each damper (kg.)	6.5 + 0.3 or - 0	4.5 + 0.3 or – 0
b) The number of dampers required per span for various span lengths and their spacing	2 for Single 4 for Twin	2

5.9.4 EARTH WIRE ACCESSORIES

5.9.4.1 MID SPAN COMPRESSION JOINT FOR GROUND WIRE:

i) The mid span compression joint for ground wire is to be used for joining of two lengths of Galvanized Steel Stranded (GSS) Ground Wire of size 7/3.15 mm for 220 kV lines and 132 kV lines.

ii) The joint sleeve shall be made of high strength hot dip galvanized steel tubes and shall be easily compressible with a 100 MT capacity hydraulic compressor. The clamp shall not permit slipping of, damage to, or failure of the complete ground wire or any part thereof at a load of not less than 95% of the ultimate tensile strength of the Ground Wire. The joint shall have conductivity more than the conductivity of an equivalent length of ground wire. The dimension and the dimensional tolerances of the joint shall be as per the following table.

Dimensions before compression			Dimensions after compression	
Inner dia (mm)	Outer dia (mm)	Length (mm)	Corner to corner width (mm)	Face to face width (mm)
10.5 +/- 0.5	21 +/- 1.0	230 +/- 5	20.2 +/- 0.5	17.5 +/- 0.5

5.9.4.2 FLEXIBLE STEEL EARTH BOND:

The flexible steel earthing bond shall be made of flexible steel cable having 34 sq. mm. copper equivalent area. The length of steel cable shall not be less than 500 mm. Two steel connecting lugs shall be press jointed to the either ends of the flexible steel cable. The complete assembly shall also include one 16mm dia 40mm long HRH M.S. bolt hot dip galvanized, with nut and lock washer.

5.10 TWIN SPACER FOR JUMPERS :

5.10.1 Jumpers at tension points shall also be fitted with spacers so as to limit the length of free conductor to 3.65m and to maintain the sub conductor spacing of 450 mm. Bidder shall quote for rigid spacer for jumper. It shall meet all the requirements of spacer used in the line except for its vibration performance. Spacers requiring retaining Rods shall not be quoted for jumpers.

5.10.2 The twin spacer offered by the Bidder shall meet the following requirements and requirements mentioned in 5.11.2

i) Any nut used, shall be locked in an approved manner to prevent vibration loosening. The ends of bolts and nuts shall be properly rounded for specified corona performance or suitably shielded.

ii) Clamp with cap shall be designed to prevent its cap from slipping out of position when being tightened.

iii) The clamp grooves shall be in uniform contact with the conductor over the entire clamping surface, except for rounded edges. The groove of the clamp body and clamp cap shall be smooth and free of projections, grit or other material, which cause damage to the conductor when the clamp is installed.

iv) Universal type bolted clamps, covering a range of conductor sizes, will not be permitted.

v) No rubbing, other than that of the conductor clamp hinges or clamp swing bolts, shall take place between any parts of the spacer. Joint incorporating a flexible medium shall be such that there is no relative slip between them.

vi) The spacer shall be suitably designed to avoid distortion or damage to the conductor or to themselves during service.

5.11 TWIN SPACER FOR POWER CONDUCTOR

5.11.1 The spacer shall be designed to maintain the bundle spacing of 450mm under all normal operating conditions and to effectively control sub span oscillation to a negligible limit. The spacer shall restore the nominal spacing after any extraordinary loads.

The spacers shall be of the armoured grip type or the bolted grip type. The spacer offered by the Bidder shall meet the following requirements and requirements mentioned in 5.11.2

i) For spacer requiring retaining rods, the retaining rods shall be designed for the specified conductor size. The rods shall be made of high strength special aluminium alloy of type 6061 or equivalent aluminium alloy having minimum tensile strength of 35 kg/mm^2. The ends of retaining rods should be ball ended. The rods shall be heat-treated to achieve specified mechanical properties and give proper resilience and retain the same during service.

ii) Four number of rods shall be applied on each clamp to hold the clamp in position. The minimum diameter of the rods shall be 7.87 +/-0.1mm and the length of the rods shall not be less than 1100mm.

iii) Where elastomer surfaced clamp grooves are used, the elastomer shall be firmly fixed to the clamp. The insert should be forged from aluminium alloy of type 6061 or equivalent aluminium alloy having minimum tensile strength of 25 kg/mm^2. The insert shall be duly heat treated and aged to retain its consistent characteristics during service.

5.11.2 a) Spacer shall restore normal spacing of the sub conductors after displacement by wind, electro-magnetic and the electrostatic forces under all operating conditions

including the specified short circuit level without permanent deformation/damage either to conductor or to the assembly itself. They shall have uniform grip on the conductors.

b) The spacer involving bolted clamps, the manufacturer must indicate the clamp bolt tightening torque to ensure that the slip strength of the clamp is maintained between 2.5kN and 5kN. The clamp when installed on the conductor shall not cause excessive stress concentration on the conductor leading to permanent deformation of the conductor strands and premature fatigue failure in operation.

c) The spacer shall not damage or chafe the conductor in any way which might affect its mechanical and fatigue strength or corona performance.

d) The clamping system shall be designed to compensate for any reduction due to conductor creep.

e) The spacer assembly shall not have any projections, cuts, abrasions etc., or chattering parts which might cause corona or RIV.

f) The spacer tube shall be made of aluminium alloy of type 6061 or equivalent aluminium alloy. If fasteners of ferrous material are used, they shall conform to and be galvanised conforming to relevant Indian Standards. The spacer involving ferrous fasteners shall not have magnetic power loss more than one watt at 600 Amps 50 Hz alternating current per sub conductor.

g) Elastomer, if used, shall be resistant to the effects of temperature upto 75 deg. C, ozone, ultraviolet radiation and other atmospheric contaminants likely to be encountered in service. It shall have good fatigue characteristics. The physical properties of the elastomer shall be of approved standard.

h) The spacer assembly shall have electrical continuity. The electrical resistance between the sub-conductor across the assembly in case of spacer having elastomer clamp grooves shall be suitably selected by the manufacturers to ensure satisfactory electrical performance and to avoid deterioration of elastomer under all service conditions.

i) The spacer assembly shall have complete ease of installation and shall be capable of removal/reinstallation without any damage.

j) The spacer assembly shall be capable of being installed and removed from the energised line by means of hot line techniques.

5.12.1 Spacer and Jumper Spacer

	Spacer	Jumper Spacer
a) Visual examination	IS:10162	IS:10162
b) Dimensional verification		
c) Chemical analysis of materials	As per Annex-C	As per Annex-C
d) Tension and compression unit	--	As per Annex-C /IS:10162
e) Clamp Slip Test	As per Annex-C /IS:10162	-do-
f) i) Vertical Vibration		--
ii) Longitudinal Vibration		--
g) Ozone test	As per Annex-C	--
h) Magnetic Power loss Test (if applicable)	As per Annex-C /IS:10162	As per Annex-C /IS:10162
i) Corona Extinction Voltage Test (dry)		-do-
j) Radio Interference Voltage Test (dry)		-do-
k) Resilience test For retaining rods for clamping		--
l) Clamp bolt torque test		--
m) Assembly torque test		--
n) Tensile load test		--

5.12.2 Armour rods for Spacers

1.	Type	Pre-formed/formed
2.	Material	Heat treated aluminium alloys as per designation HE-30 specified in IS-733
3.	Minimum ultimate strength (kg/sq. mm.)	35
4.	Conductivity	Not less than 40% of IACS
5.	Direction of lay	Right hand
6.	Shape of end	Parrot bill
7.	No. of armour rods in each set	Twelve (12)
8.	Radio Interference at phase to ground voltage of 305kV (rms) dry and frequency 1 MHz	1000 micro volts
9.	Corona extinction voltage (kV) (rms)	320
10.	Marking	Centre line to be marked with indelible black paint.

5.12.3 Spacer and Jumper Spacer for 'Moose' ACSR Conductor

1	Type	Twin Bundle
2	Sub-conductor spacing	450 mm
3	Arrangement of sub-conductors in the bundle	Horizontal
4	Galvanisation	Electro-galvanised spring washers. Other ferrous parts Hot-dip galvanised
5	Elastomers	Elastomers, if used, shall have good mechanical strength, adequate resistance to the effects of ozone, ultraviolet radiation, conductor temperature upto 75 deg. C. and atmospheric contamination under the service conditions. The elastomers shall be electrically conducting.
6	Clamp bolt strength	The clamp bolts shall be capable of withstanding a torque upto 150 percent of the recommended bolt torque without failure of component parts of the spacer/spacer damper assembly.
7	Short-circuit rating kA	50
8	Equivalent compressive and tensile strength requirements	1500 kg (compressive) 750 kg (tensile)
9	Slip strength of clamps (Both for bolted and rubber spacer type when untested)	
	i) Not less than 255 kg clamps is installed at the recommended bolt torque on the 'Moose' conductor. *	
	ii) Not less than 200 kg without re-tightening the bolts, after conducting the fatigue test.	
10	Radio Interference voltage at phase to ground Voltage of 305 kV(rms)dry and frequency 1MHz	1000 micro volts
11	Corona extinction voltage kV (rms)	320

Additional feature for spacers : -

12	Mechanical Movements	
	i) Longitudinal parallel to conductors	+/- 50 mm (minimum)
	ii) Vertical perpendicular to conductors	+/- 25 mm (Minimum)
	iii) Torsional angular movement in a vertical plane parallel to the conductors	+/- 5 deg. (minimum)

* (Both for bolted and rubber spacer/ spacer damper surfaced type when untested)

6.1 FASTENERS :

6.1.1 All fasteners shall conform to IS:6639. All fasteners shall be hot dip galvanised. All bolts and nuts shall have hexagonal heads, the heads being forged out of solid truly concentric, and square with the shank, which must be perfectly straight.

6.1.2 Fully threaded bolt shall not be used. The length of the bolt shall be such that the threaded portion shall not extend into the place of contact of the component parts.

6.1.3 All bolts shall be threaded to take the full depth of the nuts and threaded enough to permit the firm gripping of the component parts but not further. It shall be

ensured that the threaded portion of the bolt protrudes not less than 5 mm and not more than 10 mm when fully tightened.

6.1.4 Flat washers and spring washers shall be provided wherever necessary and shall be of positive lock type. Thickness of washers shall conform to IS:2016.

6.1.5 To obviate bending stress in bolt, it shall not connect aggregate thickness more than three times its diameter.

6.1.6 Bolts may have either rolled or cut threads. Nuts may be threaded after galvanising to ensure clean threads but bolts shall not be threaded or rethreaded after galvanising. Nut threads shall be tapped oversize to closely fit those of the galvanised bolt, with no unnecessary looseness but free enough to permit the nut to be turned freely with fingers over the entire threaded length. All the Bolts shall be provided with check nuts(extra nut).

6.2 FORGED COMPONENTS :

Forged components other than those covered in the clauses mentioned herein above shall be made out of class IV steel as per IS:2004 or equivalent. The items shall be normalised to achieve the required tensile properties. The forged components shall not have sharp corners and edges so as to limit corona and RIV performance.

6.3 MATERIALS :

The materials of the various components shall be as specified in clauses mentioned herein above. However, components made of alternative materials giving equivalent or better performance shall also be considered. The Bidder shall indicate the material proposed to be used for each and every component of Hardware fittings and Accessories, stating clearly the class, grade or alloy designation of the material and the reference standards. However the critical dimensions of components shall not be changed in relation to selection of different material.

6.4 GALVANISING:

6.4.1 All ferrous parts including steel components, bolts, nuts and washers etc, shall be hot dip galvanised in accordance with IS:2629 and shall satisfy the tests mentioned in IS:2633. No material shall be galvanised until all shop operations have been completed except that the nuts may be threaded and oiled after galvanising. Minimum weight of zinc coating shall be 300 g/sq.m. for fasteners and 600g/sq.m. for all other hot dip galvanised articles.

6.4.2 Fasteners shall withstand four dips while spring washers shall withstand three dips each of one minute duration in the standard preece test. Other galvanised material shall withstand minimum six dips each of one minute duration in the standard preece test for Galvanising.

6.4.3 The zinc coating shall be perfectly adherent, of uniform thickness, smooth, reasonably bright, continuous and free from imperfections such as flux, ash, rust stains, bulky white deposits and blisters.

6.4.4 The zinc used for galvanising shall be grade Zn 99.95% as per IS:209

6.4.5 The brief details of methods of Galvanising and the IS to be followed are given below.

Spring washers	- Electro galvanised as per IS 1573
Other ferrous parts	- Hot dip galvanised as per IS-2629 and IS 2633
Casting	- As per IS 729
Wires	- As per IS 326
Bolts & Nuts	- As per IS 5358

6.5 WORKMANSHIP :

6.5.1 All the material shall conform to the best modern practices adopted in the Extra High Voltage field. The Bidder shall offer only such material as guaranteed by him to be satisfactory and suitable for 220 kV and 132 kV transmission lines and Earth wire and will give continued good performance.

6.5.2 The manufacturing process and quality control of all the materials shall be such as to give maximum possible working load, highest mobility, elimination of sharp edges and corners to limit corona and radio-interference, best resistance to corrosion and a good finish.

6.5.3 Castings shall be uniform without sharp edges or corners, free from all defects like shrinkage, inclusion, blow holes, cracks etc. Pressure die casting shall not be used for casting of components with thickness more than 5 mm.

6.5.4 Forgings shall be uniform in quality and condition, without sharp edges or corners, free from cracks, flakes or seams, laps, slag, silvers, scabs, rolled-in scale and fissures.

6.5.5 All current carrying parts shall be so designed and manufactured that contact resistance is reduced to minimum.

6.5.6 No item shall have sharp ends or edges, abrasions or projections and cause any damage to the conductor in any way during erection or during continuous operation, which would produce high electrical and mechanical stresses in normal working. The design of adjacent metal parts and mating surfaces shall be such as to prevent damage of the contact surface and to maintain good electrical contact under service conditions.

6.5.7 All fasteners shall have suitable corona free locking arrangement and shall not loosen under vibration conditions.

6.5.8 Welds shall be clean, sound, smooth uniform without overlaps, properly fused and completely sealed. There shall be no cracks, voids, incomplete penetration, incomplete fusions, undercutting or inclusions. Porosity shall be minimised so that mechanical properties of the aluminium alloys are not affected. All welds shall be properly finished as per good engineering practices.

6.6 MARKINGS :

Each component of hardware shall be marked with the trade mark of the manufacturer. Marks shall be forged or stamped with a steel die before galvanising. The marks shall be distinct, durable, and conspicuous.

6.7 COMPRESSION MARKINGS:

Die compression area shall be clearly marked on components, which are to be pressure compressed such as mid span compression joints and tension hard ware. The marking shall be an etched inscription with the words **"COMPRESSION FIRST"** on components which are meant for continuous die compression. Where components are designed for intermittent die compr3ession, the compression and non-compression zones shall bear etched inscription such as **"COMPRESSION ZONE"** and **"NON COMPRESSION ZONE"** distinctly with etched arrow signs indicating the direction of compression and knurled marks showing the end of the zones.

i) All inscriptions etched on metal shall be distinct, legible and shall not wear out in the course of normal handling.

6.8 INTERCHANGEBILITY:

All components of like design shall be interchangeable.

6.9 GUARANTEED TECHNICAL PARTICULARS:

The Guaranteed Technical Particulars for the Materials being supplied shall be provided with the Bid as specified in this Specification. The Bids without the Guaranteed Technical Particulars are liable for Rejection.

7.1 TESTS :

(i) ACCEPTANCE TESTS :

These are those tests which are to be carried out during inspection prior to despatch for the purpose of acceptance of that lot.

(ii) ROUTINE TESTS :

These are those tests which are to be carried out on each Hardware fitting and Accessories to check the requirements which are likely to vary during production.

(iii) STAGE TESTS :

Stage tests during manufacture shall mean those tests which are to be carried out during the process of manufacture and end inspection by the supplier to ensure the desired quality of the end product.

7.2 ACCEPTANCE TESTS/SAMPLE TESTS:

7.2.1 On both suspension and tension hardware fittings :

a)	Visual examination	: IS:2486 (Part-I)
b)	Verification of dimensions	: IS:2486 (Part-I)
c)	Galvanising test/Electroplating	} As per Annexure of this Specification
d)	Mechanical strength test of welded joint	}
e)	Test on locking device for ball and socket coupling	} IEC:372(2) }

7.2.2 On Suspension Hardware Fittings only :

a)	Clamp slip strength Vs torque test for suspension clamp	} } As per of annexure-III of the
b)	Shore hardness test of elastomer cushion for AG suspension clamp	} specification }
c)	Bend test for armour rod set	}
d)	Resilience test for armour rods set	} IS:2121 (Part-I)
e)	Conductivity test for armour rods set	}

7.2.3 On tension Hardware Fittings only :

a) Slip strength test for dead-end assembly } IS:2486 (Part-I)

7.2.4 On Mid Span Compression Joint For Conductor And Earth Wire:

i)	Visual examination	IS:2121
ii)	Dimensional verification	IS:2121
iii)	Galvanising test	IS:2633 & IS:6745
iv)	Hardness test	
v)	Failing load test (test to be conducted after 24 Hrs. of compression)	

7.2.5 On Repair Sleeves For Power Conductor:

i)	Visual examination	IS:2121
ii)	Dimensional verification	IS:2121
iii)	Electrical Resistance Test	

7.2.6 On Flexible Steel Bond:

i)	Visual examination	IS:2121
ii)	Dimensional verification	
ii)	Slip strength test	

7.2.7 On Vibration Damper For Power Conductor

i)	Visual examination	IS:9708
ii)	Dimensional verification	IS:9708
iii)	Galvanising test	IS:9708
iv)	Verification of Resonance frequencies	
v)	Clamp slip test	
vi)	Clamp bolt torque test	
vii)	Strength of the messenger cable	
viii)	Mass pull off test	
ix)	Fatigue Test	IS-9708

7.2.8 On Spacers

i) Vibration test for spacers
ii) Clamp Slip tests
iii) Fault current test
iv) Log discrimination test
v) Dynamic characteristic test
vi) Fatigue test
vii) Ozone test
viii) Chemical analysis test
ix) Magnetic power loss test for line spacers
x) Movement test
xi) Compressive & tensile test
xii) Clamp bolt torque test
xiii) Assembly torque test
xiv) Hardness test for elastomer
xii) UTS of retaining rods

8.0 ROUTINE TESTS :

The following common tests shall be carried out on all of the Hard Ware and Accessories during course of manufacturing.

1)	Visual examination as per	IS:2121 Part-II & III, IS:2486 (Part-I)
2)	Dimensional verification as per	IS:2121 Part-II & III
3)	Proof Load Test	
4)	On Neoprene	i) Ozone test ii) Heat Resistance test

9.0 STAGE TESTS: Tests during manufacture shall be as follows:

i) **CHEMICAL ANALYSIS OF ZINC USED FOR GALVANISING**
Sample taken from the zinc ingots shall be chemically analysed as per IS:209

ii) **TESTS ON MALLEABLE CASTINGS, FORGINGS AND FABRICATED ACCESSORIES** :

These shall include chemical analysis, mechanical and metallographic tests and magnetic particle inspection for malleable castings, chemical analysis, hardness test and magnetic particle inspection for forgings and chemical analysis and mechanical tests for fabricated hardware. These shall be as per internationally recognised procedures for these tests. The samplings will be based on heat number and heat treatment batch. The details regarding these tests will be as discussed and mutually agreed to between the supplier and the purchaser in the quality assurance Programme.

10.0 TESTS DURING MANUFACTURE :

On all components as applicable
a) Chemical analysis of Zinc used for galvanising
b) Chemical analysis, hardness tests, grain size, inclusion rating and magnetic particle inspection for forgings/castings
c) Chemical analysis, and proof load test for fabricated hardware

11.0 ADDITIONAL TESTS :

a) The Purchaser reserves the right of having at his own expenses any other test(s) of reasonable nature carried out at Supplier's premises, at site, or in any other place in addition to the aforesaid type, acceptance and routine tests to satisfy that the material comply with the specifications.

b) The Purchaser also reserves the right to conduct all the tests mentioned in this specification at his own expense on the samples drawn from the site at Supplier's premises or at any other test centre. In case of any failure, it shall be binding on the part of the Supplier to replace that particular lot completely, without any extra cost to the Purchaser.

12.0 **TESTING VALUES :**

For all acceptance tests, the acceptance values shall be those values guaranteed by the Supplier in the Guaranteed Technical particulars or the acceptance values specified in the relevant standard whichever is more stringent for that particular test. The norms and procedures of sampling for these tests will be as per the Quality Assurance Programme to be mutually agreed to by the Contractor and the Purchaser. The standards and norms to which these tests will be carried out are listed against them. Where a particular test is a specific requirement of this specification, the norms and procedures of these shall be as specified in Annexure-B of this section or as mutually agreed to between the Supplier and the Purchaser in the Quality Assurance Program.

13.0 TEST SCHEDULE :

13.1 ACCEPTANCE AND ROUTINE TESTS:

The Supplier in the presence of Purchaser’s representative shall carry out all acceptance tests as stipulated in the relevant standards. The reports of the routine tests and stage tests and tests on raw materials are to be shown to the purchasers representatives at the time of inspection.

13.2 In case of any failure during testing, the cause of failure shall be thoroughly investigated by the Supplier in association with Purchaser's engineers. In case it is established that the cause of failure is due to some defect in manufacturing process or in the material itself, then the retesting shall be carried out by the Supplier without any extra cost to Purchaser. However, if it is established that the failure is due to deficiency in design itself, then the design will be suitably modified by the Supplier. After incorporating the modification suggested, retesting is to be carried out. In the latter case, the retesting charges shall be borne by the supplier. No delivery extension shall be given for this additional testing. In case of any dispute, the assessment of the Purchaser as to what is the cause of failure shall be final and binding.

13.3 The entire cost of testing for acceptance and routine tests and tests during manufacture specified herein shall be treated as included in the quoted unit price of materials, except for the expenses of the inspector/Purchaser's representative.

13.4 Immediately after finalization of the program of acceptance testing, the Supplier shall give sufficient advance intimation to the Purchaser, to enable him to depute his representative for witnessing the tests.

14.0 INSPECTION:

14.1 The Purchaser's representative shall, at all times, be entitled to have access to the works and all other places of manufacture, where the hardware fittings, Accessories and its components are being manufactured and the Supplier shall provide all facilities for unrestricted inspection of the Bidder's works, raw materials, process of manufacture of all the accessories and for conducting necessary tests as detailed herein.

14.2 The Supplier shall keep the Purchaser informed in advance of the time of starting and of the progress of manufacture of accessories in its various stages so that arrangements could be made for inspection.

14.3 The supplier shall give 15 days advance, intimation to enable the purchaser to depute his representative for witnessing acceptance and routine tests.

14.4 No material shall be despatched from its point of manufacture before it has been satisfactorily inspected and tested, unless the inspection is waived off by the Purchaser in writing. In the later case also, the Materials shall be dispatched only after satisfactory testing for all tests specified herein has been completed and approved by the Purchaser.

14.5 The Acceptance tests shall be carried out on all the materials offered, by selecting the samples as per relevant IS specified, at random. If any sample drawn does not pass the acceptance tests, the sample will be rejected and another sample of the same material from same lot shall be selected at random to repeat the "Acceptance tests". If the second sample also fails in the same Test, the entire materials of that item in that lot offered for Inspection will be rejected.

14.6 The acceptance of any quantity of accessories shall in no way relieve the Supplier of his responsibility for meeting all the requirements of this specification and shall not prevent subsequent rejection if such hardware fittings are later found to be defective.

15.0 DOCUMENTATION:

i) The Bidder shall furnish along with the Bid the dimensional drawings of all types of hardware fittings.

ii) These drawings shall include the following information.

a) Dimensions
b) Tolerances on dimensions
c) Material designation used for different components with reference to standards
d) Fabrication details such as welds, finishes and coatings
e) Catalogue or part numbers for each component and the total assembly with bill of materials
f) Identification marking
g) Weight of individual components and total assembled weight

iii) The assembly drawings shall include the following :

a) Brief installation instructions
b) Design installation torque for the bolt or cap screw
c) Withstand torque that can be applied to the bolt or cap screw without failure of components or parts thereof
d) Compression die number with recommended compression pressure
e) Relevant technical details of significance

16.0 TEST REPORTS:

i) Three copies of acceptance test reports shall be furnished to the purchaser for approval and only there after shall the materials be dispatched.

ii) All record of routine test reports shall be maintained by the Supplier at his works for inspection at the time of acceptance tests by the Purchaser's representative.

iii) The Supplier shall maintain all test reports of tests conducted during manufacture. These shall be produced for verification as and when requested for by the Purchaser.

17.0 PACKING AND FORWARDING :

i) All hardware fittings shall be packed in suitably sized, strong and weather resistant wooden cases/crates. The gross weight of the packing shall not normally exceed 200 kg. to avoid handling problems.

ii) Suitable cushioning, protective padding, or spacers shall be provided to prevent damage to or deformation of the hardware during transit and handling.

iii) All identical items shall be despatched to destination duly assembled and packed. Bolts, Nuts, Washers, Cotter pins, security clips and split pins etc. shall be packed duly installed and assembled with the respective parts and suitable measures shall be taken to prevent their transit loss.

iv) Each component part shall be legibly and indelibly marked with the trade mark of the manufacturer and year of manufacture.

v) All packing cases shall be marked legibly and correctly so as to ensure safe arrival at their destination and avoid the possibility of goods being lost or wrongly despatched on account of faulty or illegible markings. Each wooden case/crate shall have all the markings stenciled on it in indelible ink.

vi) The scope of supply shall include supply of 2.5% extra quantity of bolts, nuts, washers, split pins, cotter pins and such other small loose items free of cost.

ANNEXURE – A

DETAILS FOR POWER CONDUCTOR

	PARAMETER	PANTHER ACSR	MOOSE ACSR
a)	Stranding and wire diameter	30/3.00 mm Aluminium 7 / 3.00 mm Steel	54/3.53 mm Aluminium 7 / 3.53 mm Steel
b)	**A. Number of Strands**		
	Central Steel wire	1	1
	1st Steel Layer	6	6
	1st Aluminium Layer	12	12
	2nd Aluminium Layer	18	18
	3rd Aluminium Layer	-	24
c)	Sectional area of Aluminium (Sq.mm)	212.10	528.50
d)	Total Sectional area (Sq.mm)	261.50	597.00
e)	Overall diameter (mm)	21.00	31.77
f)	Approximate weight (kg/km)	974	1998

g)	Calculated maximum D.C resistance at 20 deg. C (Ohm/km)	0.14	0.05595
h)	Minimum UTS (kN)	89.67	159.60
i)	Modulus of Elasticity (gN/Sq.meter)	80	69
j)	Coefficient of linear expansion per deg C	17.80×10^{-6}	19.30×10^{-6}

B) DETAILS FOR HTGS WIRE

1.	Galvanised standard steel wire	:	7/3.15 mm Stranded High Tensile Galvanised Steel wire.
2.	Stranding and wire diameter	:	7/3.15 mm
3.	Standard weight	:	432 Kg/Km
4.	Single wire before stranding	:	
	i) Diameter of wire	:	3.15 mm
	ii) Tolerance	:	+ 0.06mm/- 0.03 mm
	iii) Minimum breaking load	:	860 Kgs.
5.	Minimum breaking load of stranded wire	:	56. 98 kN.
6.	Minimum ultimate tensile strength	:	109 Kgf/mm Sqr.
7.	Overall diameter of stranded wire	:	9.45 mm

(C) DETAILS OF INSULATORS

Sl. No.	Description		Electrical / Mechanical
1.	Type of Insulator.	:	(Ball & Socket) Disc Insulators (70KN, 120KN 160KN)
2.	Size and designation of Ball & Socket with standard	:	Refer 5.3
		:	IS-2486 Part – II to which it will Confirm.

3.	Dimensions		70KN	120KN	160KN
	a) Porcelain disc diameter in mm	:	255	280	280
	b) Unit spacing in mm	:	145	145	170

4. Mechanical values

a) Combined Mechanical and Electrical strength in Kgs : :

	132kV	220kV	220kV (Twin Moose)
Tension	11500	11500	16500
Suspension	7000	7000	11500

b) Routine Mechanical test load in Kgs : 2,800 Kgs

5.	Net weight of each Insulator - Kgs	:	5 Kgs (Approx.)
6.	Standard specification to which insulators conform	:	IS-731-1971
7.	Number of disc insulators per String	:	9 & 10 for 132 kV and 13 & 14 for 220 kV Tension Suspension & respectively.

ANNEXURE-B

DETAILS OF TESTS (Hardware Fittings)

1. GALVANISING/ELECTROPLATING TEST :

The test shall be carried out as per clause no.5.9 of IS:2496 (Part-I) except that both uniformity of zinc coating and standard Preece test shall be carried out and the results obtained shall satisfy the requirements of this specification.

2. MECHANICAL STRENGTH TEST FOR WELDED JOINT :

The welded portion of the component shall be subjected to a load of 2000 kgs. for one minute. Thereafter, it shall be subjected to die-penetration/ultrasonic test. There shall not be any crack at the welded portion.

3. CLAMP SLIP STRENGTH VS TORQUE TEST AND CLAMP SLIP FOR AG SUSPENSION CLAMP/ENVELOPE TYPE SUSPENSION CLAMP :

The suspension clamp shall be vertically suspended by means of a flexible attachment. A suitable length of ACSR conductor shall be fixed in the clamp. The clamp slip strength at various tightening torques shall be obtained by gradually applying the load at one end of the conductor. The clamp slip strength vs. torque curve shall be drawn. The above procedure is applicable only for free centre type suspension clamp. For AG suspension clamp only clamp slip strength after assembly shall be found out.

4. SHORE HARDNESS TEST OF ELASTOMER CUSHION FOR AG SUSPENSION ASSEMBLY :

The shore hardness at various points on the surface of the elastomer cushion shall be measured by a shore hardness metre and the shore hardness number shall lie between 65 to 80.

5. PROOF LOAD TEST :

Each component shall be subjected to a load equal to 50% of the specified minimum tensile strength. The load shall be held for one minute and then removed. After removal of the load the component shall not show any failure or permanent deformation.

6. CHEMICAL ANALYSIS OF ZINC USED FOR GALVANISING :

Samples taken from the zinc ingot shall be chemically analysed as per IS:209. The purity of Zinc shall not be less than 99.95%.

7. TEST FOR FORGING, CASTING AND FABRICATED HARDWARE :

The chemical analysis, hardness test, grain size, inclusion rating and magnetic particle inspection for forging, casting, radio graphical inspection for items made of aluminium alloys, chemical analysis and proof load test for fabricated hardware and ductility test on tensile test specimen shall be as per the internationally recognised procedures for these tests. The sampling will be based on heat number and heat treatment batch. The details regarding test and acceptance criteria will be as per the Quality Assurance Programme agreed upon.

8. TESTS ON NEOPRENE :

a) Ozone Test :

This test shall be performed in accordance with ASTM-D-1171 by the ozone chamber exposure method (Method-B). The test duration shall be 500 hrs. and the ozone concentration should be 50 PPM. At the test completion, there shall be no visible crack under a 2X magnification.

b) Heat Resistance Test :

This test shall be performed in accordance with ASTM-D-573. The sample shall be kept in an oven for accelerated ageing for 70 hrs. at 100 Deg. C. After the test, the sample shall be tested for tensile strength, elongation and hardness tests.

9. Magnetic Power loss test for the suspension clamps:

The test may be conducted on an actual conductor fitted with armour rods of suitable length by differential power loss method making use of low PF wattmeter, current transformer without phase shift and fixed gain differential amplifier.

In this method a Wheat-Stone bridge is formed by the conductor and two equal resistances included in the circuit. By using a slider contact on the conductor, the power loss in both the arms of the conductor can be made equal. Thus initially the bridge is balanced. Introduction of the clamp shifts the balance and shift is proportional to the power loss introduced by the clamp. The balance circuit shall be crosschecked by loading a known resistance through a CT in one of the arms.

The test shall be carried out on individual clamp and values on 3 clamps shall be recorded. The test shall be conducted at 200, 400,600 and 800 Amps. in the case of each clamp. The power loss of each clamp at 600 amps. Shall be recorded and the power loss of any clamp shall not be more than 13 W. A graph shall be drawn based on the average value of each clamp at 200, 400, 600 and 800 Amps. The value at 600 Amps. Taken from the graph shall not exceed 12 watts.

NOTE: The clamp under test shall consist of components including bolts up to and excluding yoke plate.

10. **Mechanical test for the complete string fittings:**

A single tension/suspension assembly consisting of respective string fittings but excluding clamps shall be mounted in a tensile testing machine. The assembly shall then be subjected to a tensile load by increasing the load from 0 to 80% of the rated strength of the fitting in a time not exceeding two minutes. This load small be maintained on the assembly for one minute after which the assembly shall be unloaded and dismantled with hand or shall hand tools and the string fittings shall be inspected for visual deformation. Should the bolts used in the assembly indicate a mid point deflection of more than one mille metre with respect to their ends or any difficulty be experienced in dismantling and re-assembling the string fittings with hand or small hand tools and/or if there is any visual deformation caused by the application of the above load, the string fittings which exhibit deformation shall be deemed to have failed in this test. If the string fittings can be re-assembled, the assembly shall be again mounted in the testing machine and loaded up to full value of the rated strength in a time not exceeding two minutes. The load on the assembly shall be continued to be raised up to 110% of the rated strength or till breakage occurs whichever is lower. The string fittings shall be deemed to have passed this test if the breakage occurs at a load not lower than their rated strength. The behavior of the string fittings at loads above their rated strength shall be for the sake of information only.

DETAILS OF TESTS (Accessories)

1. Mid Span Compression Joints For Conductor And Earth Wire :

a) Failing Load Test :

The fitting compressed on conductor/ground wire shall not be less than one metre in length. The test shall be carried out as per IS:2121(Part-II) except that the load shall be steadily increased to 95% of the minimum ultimate tensile strength of conductor/ground wire and retained for one minute at this load. There shall be no

movement of the conductor/ground wire relative to the fittings during this one minute period and no failure of the fittings.

b) Hardness Test: As per the values guaranteed by the Bidder

2. FLEXIBLE STEEL BOND :

a) Slip strength Test :

On applying a pull of 3 kN (300 kg) between the two ends, stranded flexible steel cable shall not come out of the connecting lugs and none of its strands shall be damaged. After the test, the lugs shall be cut open to ascertain that the gripping of cable has not been affected.

3. VIBRATION DAMPER FOR CONDUCTOR :

a) Dynamic Characteristics Test :

The damper shall be mounted with its clamp tightened at the torque recommended by the manufacturer on shaker table capable of simulating sinusoidal vibrations for frequencies ranging from 5 to 45 Hz. The damper assembly shall be vibrated vertically with a double amplitude of 1 mm at different frequencies to determine the following characteristics with the help of suitable recording instruments.

i) Force Vs Frequency
ii) Phase angle Vs Frequency
iii) Power dissipation Vs Frequency.

The above characteristics shall be determined by averaging the characteristics of the damper by carrying out this test five times on a single damper.

b) Vibration Analysis :

The vibration analysis shall be done on a computer. The following parameters shall be taken into account for the purpose of analysis

i) The analysis shall be done for a single conductor (28.62 mm)) without armour rods as per the parameters given in Clause 5.5 (vii) for power conductor. The tension shall be taken as 44.2 kN (4500 kg) for power conductor for a span ranging from 250 m. to 440 m.

ii) The self damping factor and flexural stiffness (EL) for power conductor shall be calculated on the basis of experimental results. The details of experimental analysis with these data should be furnished.

iii) The power dissipation curve obtained from dynamic Characteristic test shall be used.

iv) Examination of the Aeolian vibration level of vibration damper for wind velocity ranging from 0 to 30 km per hour, predicting amplitude frequency and vibration energy input.

c) Clamp slip and fatigue tests :

i) **Test Set Up :** The clamp slip and fatigue tests shall be conducted on a laboratory set up with a minimum span length of 30M. The power conductor shall be tensioned at 44.2KN (4500 kg/ 14.72 kN (1500 kg) and shall not be equipped with protective armour at any point. After the power conductor has been tensioned, a clamp shall be installed to rigidly support the conductor at the end where the measurements shall be taken. There shall be no loose parts, such as Suspension Clamp, U-bolts on the test span. The span shall be equipped with vibration inducing equipment suitable for producing steady standing vibration. The equipment shall be suitably for measuring the amplitude of vibration at any point along the span.

ii) **Clamp Slip Test :** The vibration damper shall be installed on the test span. The damper clamp, after tightening with the manufacturer's specified torque, when subjected to a longitudinal pull of 2.5 kN (250 kg) parallel to the axis of power conductor shall not slip i.e., the permanent displacement between conductor and clamp measured after removal of the load shall not exceed 1.0 mm.

iii) **Fatigue Test :** The vibration damper shall be installed on the test span with the manufacturer's specified torque. The damper shall then be vertically vibrated sinusoidally at its highest resonant frequency as determined by the characteristic of the particular sample so as to give an amplitude of + 1 mm for power conductor damper at the damper clamp location for ten million cycles. The test shall be done separately for the highest resonant frequencies of each damper mass in case of 4R damper. The clamp slip test as mentioned herein above shall be repeated without retorquing the damper clamp and shall withstand a minimum load of 2 kN (200kg.).

After the above tests, the damper shall be removed and subjected to dynamic characteristic test. There shall not be any deterioration in the characteristic of the damper nor shall there be any broken, loose or damaged parts. There shall not be any significant deterioration or wear of the damper and the power conductor under the clamp shall also be free from any damage.

iv) Verification of Resonance Frequencies:

The damper shall be mounted on a shaker table and vibrated to determine the resonant frequencies of the damper. The resonant frequencies obtained shall be compared with the resonant frequencies from the dynamic characteristic curve obtained during type tests with a tolerance of + 1 Hz at a frequency lower than 15 Hz and + 2 Hz at a frequency higher than 15 Hz.

v) Clamp Bolt Torque Test :

The clamp shall be attached to a section of the power conductor. A torque of 150 percent of the manufacturer's specified torque shall be applied to the bolt. There shall be no failure of component parts. The test set up shall be as described in Clause 3 (c) (i) Annexure-III(a).

vi) Strength of the Messenger Cable:

The messenger cable sample shall be fixed in a suitable tensile testing machine and a tensile load shall be gradually applied until yield point is reached. The load shall be not less than the value guaranteed by the Supplier.

vii) Mass Pull off Test :

Each mass shall be pulled off in turn by fixing the mass in one jaw and the clamp in the other of a suitable tensile testing machine. The longitudinal pull shall be applied gradually until the mass begins to pull out of the messenger cable. The pull off loads shall not be less than the values guaranteed by the supplier.

4. VISUAL EXAMINATION : (IS:2121)

All parts shall be checked visually for good workmanship and smooth finish.

5. DIMENSIONAL VERIFICATION : (IS:2121)

All dimensions shall be checked against approved drawings.

6 MAGNETIC POWER LOSS TEST:

The vibration damper shall be tested for magnetic loss simulating the service conditions for pure sine wave, power loss shall be determined by averaging the losses of four dampers. The damper shall not have a magnetic power loss more than 1.0 watt.

7. DAMPING EFFICIENCY TEST :

This test should be carried out in accordance with IS -9708 and determination of the damping efficiency done by suitably evaluating the amplitude of the conductor at the anti-nodal and nodal points under actual span vibrations. For the purpose of this test an actual tensioned span of not less than 35 meters should be used.

8. BENDING STRAIN TEST :

It is intended that the efficiency of the damper may be proved by vibration tests on and actual conductor span by showing that i) maximum bending strain does not exceed the prescribed limit and ii) over the extended vibration tests upto 10 million cycles, the conductor does not offer any fatigue such as to cause the breaking of the strands. The Bidder shall provide a computer analysis by a mathematical model of calculated bending strain for conductor with specified parameters for the full range of wind velocities, frequencies of vibration, amplitude of vibration and loop length. Such calculations shall be made available for the undamped conductor. The vibration analysis shall be conducted with the damper on Moose ACSR conductor. Actual tests on a test span with a suitable vibration facility shall then be carried out to show that the bending strain does not exceed the prescribed limit and secondly over the extended period of vibration, the damper is adequately effective to prevent objectionable fatigue in the conductor. The vibration dampers shall conform to IS-9708-1980.

ANNEXURE-C

1.0 Tests for Spacers:-

a) Vibration Tests for Spacer

The test set up shall be as per clause no. 3(c) (i) above. The spacer assembly shall be clamped to conductor. During the vibration tests the axis of the clamp of sample shall be maintained parallel to its initial static position by applying a tension of 45 KN on the conductor. The spacer assembly shall be free to vibrate and shall not be re-torqued or adjusted between the tests.

All the vibration tests mentioned hereunder shall be conducted on the same sample and the same test span. The samples shall withstand the vibration tests without slipping on the conductor, loosening, damage or failure of component parts. After each vibration test clamp slip test shall be carried out as per the procedure given in clause no.4 (b)

i) **Longitudinal Vibration Test**

The stationary conductor shall be restrained and the vibrating conductor/equivalent diameter of aluminium alloy tube shall be restrained by fixed clamps. The displacement of the vibrating conductor shall be 25 mm on either side. The longitudinal movement shall be parallel to the conductor at frequency not less than 2Hz for minimum one million cycles.

ii) **Vertical Vibration Test**

The spacer damper shall be installed in the middle of the test span and the frequency chosen so as to get an odd number of loops. The shaker shall be positioned at least two loops away from the test specimen to allow free movement of the conductor close to the test specimen. One conductor shall be connected to the shaker and vibrated to an amplitude such that.

$$f.Ymax > 120 \text{ mm/Sec}$$

Where Ymax being the anti node amplitude (mm) and f is the test frequency shall be greater than 24 Hz and the total number of cycles shall be more than 10 millions.

iii) **Sub span Oscillation Test**

The test shall be conducted for oscillation in horizontal plane at frequency higher than 3 Hz for minimum one million cycles. The amplitude of oscillation shall be kept equivalent to an amplitude of 150 mm for a full sub-span of 80m. Both the conductors shall be vibrated 180 deg. out of phase with the above minimum amplitude.

b) Clamp Slip Test

The spacer assembly shall be installed on test span of twin conductor bundle string at a tension of 45 kN (4500 kg). In case of spacer/spacer damper, the clamp of sample shall be tightened with a specified tightening torque. One of the clamp of the sample when subjected to a longitudinal pull of 2.5 kN (250 kg) parallel to the axis of conductor for a minimum duration of one minute, shall not slip on the conductor i.e, the permanent displacement between the conductor and the clamp of sample measured after removal of the load, shall not exceed 1.0 mm. Similar test shall be performed on the other clamp of the same sample. Such clamp slip tests shall also be conducted after each of the vibration test mentioned in Clause 4 (a). Each clamp shall withstand a minimum longitudinal load of 2 kN (200 kg) for a minimum duration of one minute after the vibration test without any adjustment of sample.

c) **Fault Current Test**

An outdoor test span of twin bundle conductor shall be strung horizontally at a tension of 45 kN (4500 kg). The spacer assembly shall be installed at recommended intervals simulating a normal sub span on the central sample (test specimen). A fault current of 40 kilo Amperes asymmetrical with 1.6 offset, shall be applied to the conductors five successive times, each time for a minimum duration of 5 cycles. The spacer assembly under test, shall withstand this test without slipping on the conductor, loosening or failure of component parts.

d) **Log Decrement Test**

The spacer assembly shall be mounted on test of twin conductor bundle at a tension of 45 kN (4500 kg). The test span shall be instrumented to continuously monitor and record the horizontal motion of the sub conductor in the subspan between suspension point and the first sample. The log decrement test shall be made with an initial peak to peak amplitude of four to six times the conductor diameter in the middle of the subspan being considered. The conductor shall be excited in a horizontal one loop per subspan resonant mode with a slow and steady build up of amplitude that minimises harmonics and other distortions. After achieving a steady state motion, the conductor excitation shall be discontinued leaving the conductor undisturbed. The motion shall be recorded until it reduces to an amplitude of half the conductor diameter. The logarithmic (log) decrement shall be the value for a minimum reduction of 80% in amplitudes. The minimum acceptable log decrement average for five or more excitations shall be 0.04 based upon the following formula for decay.

$$= \text{Log}_e \frac{(An)}{(An+1)} = \frac{1}{n} \text{Log}_e \frac{(Ao)}{(An)}$$

Where, Ao is the initial amplitude and An is the amplitude n cycles later.

e) **Dynamic Characteristic Test**

The purpose of this test is to obtain quantitative information regarding the dynamic characteristics of the spacer. The values obtained during this test will serve as reference to evaluate the behaviour of the same spacer under the fatigue test.

The test will consist in the application of sinusoidal movement to the spacer articulation and measuring the force (F), displacement (X) and phase angle (ϕ) between these two. From these values, the stiffness (K_ϕ) and the damping factor (n) will be calculated.

$$K_\phi = \frac{F \cos \phi}{X} ; \qquad n = \tan\phi$$

The test frequency shall not be higher than 3 Hz. The test shall be performed at five different displacement amplitudes. The amplitudes shall be selected to reproduce 10, 20, 40, 60 and 90 percent of the maximum displacement permitted by the spacer design.

The test shall be performed on three samples.

f) **Fatigue Test**

The purpose of this test is to evaluate the capacity of the spacer to sustain without damage the cyclic movements which can be induced by vibrations.

The spacer articulation shall be subjected to cyclic motions for a total of 1 million cycles. The test frequency shall be between 2 and 3 Hz. The amplitude of motion shall be established on the following basis :

- the load applied on the spacer-damper clamp shall not be less than +/-300 N.

- the clamp displacement under the applied load shall not be less than 60% of the maximum displacement permitted by the design.

- if the 300 N load generates movement exceeding the maximum permitted displacement, the load can be reduced to limit the movement to 95% of the maximum displacement.

After the test, the sample shall be subjected to a second dynamic characteristic test. This test shall be performed at two amplitudes, 10% and 60% of the maximum displacement.

The spacer shall show no signs of cracks or deterioration, loosening of bolts or abnormal wear.

The dynamic characteristics (k and n) shall not be less than 60% of the values measured before the fatigue test. The test shall be performed on three samples.

g) **Ozone test**

This test shall be performed in accordance with ASTM D - 1171 by the ozone chamber exposure method (method B). The test duration shall be 500 hours and the ozone concentration 50 PPHM. At the test completion, there shall be no visible crack under a 2x magnification.

h) **Chemical Analysis Test**

Chemical analysis of the material used for manufacture of items shall be conducted to check the conformity of the same with technical specification and approved drawing.

i) **Magnetic Power Loss Test for Spacers for line**

The sample involving ferrous parts shall be tested in a manner to simulate service conditions for 50 Hz pure sine-wave. The test should be carried out at various currents ranging from 300 amperes to 900 amperes and the magnetic power loss at various currents should be specified in tabulated graphical form. The difference between the power losses without and with sample at room temperature shall be limited to 1 watt for 600 amperes current (rms). The losses shall be determined by averaging the observations obtained from at least four samples.

2.0 **Common Tests for Spacer/Jumper spacer**

a) **Test Set up**

The test set up for the tests described hereunder is as per clause 1.0 (c) of this Annexure.

b) **Movement Test**

The spacer/jumper spacer assembly shall be capable of the following movements without damaging the conductor, assuming one conductor is fixed and the other moving:

i) Longitudinal movement +/- 50 mm parallel to the conductor.

ii) Vertical movement in a +/- 25 mm vertical direction at right angle to the conductor.

iii) Torsional movement/angular +/- 5 deg. movement in a vertical plane parallel to the conductor.

c) **Compressive and Tensile Test**

The spacer/jumper spacer assembly shall withstand ultimate compressive load of 15 kN (1500 kg) and tensile load of 7.5 kN (750 kg) applied between sub-conductor bundle and held for one minute without failure. Line distance between clamps shall be recorded during each of the compression and tension test. Measurement shall be recorded at (i) no load (ii) with load (iii) after release of load.

The centre line distance under load shall be within +100 mm of the nominal design spacing. After release of load it shall be possible to retain the clamps at their original position using only slight hand pressure. There shall be no deformation or damage to the spacer assembly which would impair its function of maintaining the normal spacing.

d) **Clamp Slip Test**

Same as clause 1.0 (b) of this Annexure.

e) **Clamp Bolt Torque Test**

The spacer/jumper spacer assembly shall be attached to conductor. A torque of 150 percent of the manufacturer's specified tightening torque shall be applied to the clamp bolts or cap screws. There shall be no failure of the component parts.

f) **Assembly Torque Test**

The spacer/jumper spacer assembly shall be installed on conductor. The same shall not rotate on either clamp on applying a torque of 0.04 kN (4kg) in clockwise or anti-clockwise direction.

g) **Hardness test for elastomer**

The shore hardness at different points on the elastomer surface of cushion grip clamp shall be measured by shore hardness meter. They shall be lie between 65 to 80.

h) **UTS of Retaining Rods**

The ultimate tensile strength of the retaining rods shall be measured. The value shall not be less than 35 kg/mm2.

GUARANTEED TECHNICAL PARTICULARS FOR SUSPENSION AND TENSION CLAMPS FOR GALVANISED STEEL EARTH WIRE

[to be filled and furnished along with the Bid (part-1)]

Sl. No	Description	Suspension Clamp	Tension Clamp
(1)	(2)	(3)	(4)
1.	Maker's name and Country		
2.	Type		
3.	Design as per Drawing No (give complete details and dimensions)		
4.	Manufactured as per ISS		
5.	Material used		
6.	Quality of material used		
7.	Make of bolt, nut & washers used		
8.	Ultimate strength of material used		
9.	Method of manufacture adopted e.g. forging or casting		
10	Standard specification to which galvanising conforms		
11	Minimum slip strength (kg)		
12	Minimum ultimate strength (kg)		
13	Tested to standard specification		
14	Total weight of each clamp		
15	Packing details		
16	Certification mark (BIS/BS/ Any national institute)		
17	Whether the materials covered by this bid have been fully type tested and if so, whether the copies of the type test certificates enclosed to the bid offer.		

GUARANTEED TECHNICAL PARTICULARS FOR MIDSPAN COMPRESSION JOINTS AND REPAIR SLEEVES FOR ACSR CONDUCTOR AND MIDSPAN COMPRESSION JOINTS FOR GALVANISED STEEL EARTHWIRE

[to be filled and furnished along with the Bid (part-1)]

Sl No	Description	For MOOSE ACSR Conductor		For PANTHER ACSR Conductor		For GS Ground-wire
		Midspan compress-ion joints	Repair Sleeves	Mid span compress-ion joints	Repair Sleeves	Mid span compression joints
(1)	(2)	(3)	(4)	(5)	(6)	(7)
1.	Maker's name & Country					
2.	Type	Compression Type				
3.	Suitable for	MOOSE ACSR		PANTHER ACSR		GS EARTH-WIRE
4.	Outer diameter of sleeves (mm)					
a)	Before compression					
i)	Aluminium					
ii)	Steel					
b)	After compression					
i)	Aluminium					
ii)	Steel					
c)	After compression					
i)	Aluminium					
ii)	Steel					
5.	Inner diameter of sleeve before compression (mm)					
i)	Aluminium					
ii)	Steel					
6.	Length of sleeve					
a)	Before compression					
i)	Aluminium					
ii)	Steel					
b)	After compression					
i)	Aluminium (Approx)					
ii)	Steel (Approx)					

7.	Wt of sleeve per joint (kg / Approx)							
i)	Aluminium							
ii)	Steel							
iii)	Total							

8.	Guaranteed breaking strength of the mid span joint expressed as percentage of ultimate tensile of Conductor / ground wire given in the specification	
9.	Conductivity of the compression joint expressed as percentage of the conductivity of the conductor / ground wire	
10	Full details of the hydraulic compressor and dies required	
11	Guaranteed slip strength of the mid span joint expressed as percentage of ultimate tensile of Conductor / ground wire given in the specification	
12	Resistance as percentage of measured resistance of equivalent length of conductor / ground wire	
13	Standard specification to which the materials will be manufactured and tested	
14	Whether the materials covered by this bid have been fully type tested and if so, whether the copies of the type test certificates enclosed to the bid offer.	

Signature of Bidder
With Seal and Date

GUARANTEED TECHNICAL PARTICULARS FOR VIBRATION DAMPERS FOR ACSR CONDUCTOR

[to be filled and furnished along with the Bid (part-1)]

Sl No	Description	For MOOSE ACSR Conductor	For PANTHER ACSR Conductor
(1)	(2)	(3)	(4)
1	Maker's name & Country		
2.	Type		
3	Suitable for (Conductor size)	MOOSE ACSR	PANTHER
4	Total weight of each damper	6.6 kg	4.6 kg
5	Diameter of balancing weights (mm)		
6.	Length of balancing weights (mm)		
7	Weight of each balancing weights (kg)		
8	Tolerance of balancing weights (%)		
9	Slip strength of steel messenger cable (kg)		
10	Diagram showing power dissipated by the damper in watts for various frequencies and amplitudes.		
11	Natural frequency of the damper		
a)	Upper (Cycles per sec)		
b)	Lower (Cycles per sec)		
12	The numbers of dampers required per span for various span lengths and their spacings	SPAN LENGTHS	
		Up to 400 m	Above 400m to 700 m
13	Test reports of the damping characteristics and energy dissipation		
14	Whether a write-up explaining the suppression of Aeolian vibration dampers is given in the bid.		
15	Damping efficiency of the damper		
16	Wind velocity range for which damper will be effective		
17	Standards to which materials will be manufactured and tested		
18	Packing details		
19	Whether the materials covered by this bid have been fully type tested and if so, whether the copies of the type test certificates enclosed to the bid offer		

Signature of Bidder
With Seal and Date

GUARANTEED TECHNICAL PARTICULARS OF SPACER

FOR MOOSE ACSR CONDUCTOR

([to be filled and furnished along with the Bid (part-1)]

1.		Maker's name & address	
2.		Type / governing standard	
3.		Suitable for (conductor size) (mention Drg. and Catalogue nos. also) (mm)	
4		Materials, grade and standards to which conforming i) Body ii) Arms iii) Bushes iv) Elastomer	
5		Total weight of each spacer (kg)	
6		Means of fixing clamps i) Bolted type/retaining rod type ii) Clamp bolt tightening torque/ number of retaining rods and type iii) Means of bolt torque control	
7		Spacing between subconductor (mm) i) Along run of line ii)At suspension insulators string iii)At tension insulator string	
8		Distance between support clamp and first spacer(mm) i) At tension position ii)At suspension position	
9		Minimum no. of spacer per span for span lengths from 100m to 1000m	
10		Maximum distance between adjacent spacers of each phase along run of line (m.)	
11		Inter conductor spacing (mm) i) Under normal full load current ii) Under fault conditions	
12		Ultimate strength of the damper (kg) i) Compressive load ii) Tensile load	
13		Movement which can be applied to the spacer i) Longitudinal parallel to conductor (mm) ii) Vertical (mm) iii)Torsional (degrees)	
14		Magnetic power loss per spacer at 600 amps. 50 cycles/ sec. AC current (watts)	

15		Corona performance	
	a)	Corona inception and extinction voltage line to earth (kV) i) Dry condition ii) Wet condition	

	b)	Corona power loss per spacer in watts at rated voltage (watts) i) Dry condition ii) Wet condition	
16		Radio Interference voltage performance i) Dry conditions at 305 kV (rms) phase to ground ii) Wet conditions corresponding to 15.	
17		Minimum fatigue strength of spacer in nos. of cycles for specified amplitudes and frequencies (kg/Hz) For sub span oscillation	
18		Clamp slipping strength of spacer clamp	
	a)	Before vibration test when tightened at specified torque	
	b)	After the vibration test i) Without re torqueing the bolts and cap screws (kg) ii) With re torqueing the bolts and cap screws (kg)	
19.		Galvanisation	
	a)	Method of galvanisation	
		i) Spring Washers	
		ii) any other spare parts	
	b)	Quality of Zinc and standard to which conforming	
20.		Short circuit Rating (kA)	
21.		Equivalent compressive and tensile strength (kg)	
22.		Electrical resistance of elastomer (ohms)	
23.		Electrical resistance between sub-conductors across the assembly (ohm)	
24.		Maximum conductor temperature which can be with stood by the elastomer (deg. C.)	
25.		Certification Mark (ISI/BS/or any other International Standard)	
26.		Packing details	

Signature of Bidder
With Seal and Date

GUARANTEED TECHNICAL PARTICULARS OF JUMPER SPACER

FOR MOOSE ACSR CONDUCTOR

[to be filled and furnished along with the Bid (part-1)]

1.		Maker's name & address	
2.		Type / governing standard	
3.		Suitable for (conductor size) (mention Drg. and Catalogue nos.	

		also) (mm)	
4		Materials, grade and standards to which conforming i) Body ii) Arms iii) Bushes iv) Elastomer	
5		Total weight of each jumper spacer (kg)	
6		Means of fixing clamps i) Bolted type ii) Clamp bolt tightening torque iii) Means of bolt torque control	
7		Distance between tension clamp lug and jumper spacer (mm) at tension position	
8		Minimum no. of jumper spacer per jumper.	
9		Maximum distance between adjacent spacers of each phase along run of jumper (m.)}	
10		Inter conductor spacing(mm) i) Under normal full load current ii) Under fault conditions	
11		Ultimate strength of the jumper spacer (kg) i) Compressive load ii) Tensile load	
12		Magnetic power loss per spacer damper at 600 amps. 50 cycles/ sec. AC current (watts)	
13		Movement which can be applied to the spacer i) Longitudinal parallel to conductor (mm) ii) Vertical (mm) iii)Torsional (degrees)	
14		Magnetic power loss per spacer at 600 amps. 50 cycles/ sec. AC current (watts)	
15		Corona performance	
	a)	Corona inception and extinction voltage line to earth (kV) i) Dry condition ii) Wet condition	
	b)	Corona power loss per spacer damper in watts at rated voltage (watts) i) Dry condition ii) Wet condition	
16		Radio Interference voltage i) Dry conditions performance at 305 kV (rms) phase to ground ii) Wet conditions corresponding to 15.	
17		Minimum fatigue strength of spacer in nos. of cycles for specified amplitudes and frequencies (kg/Hz)	

		For sub span oscillation	
18		Clamp slipping strength of spacer clamp	
	a)	Before vibration test when tightened at specified torque	
	b)	After the vibration test i) Without re torqueing the bolts and cap screws (kg) ii) With re torqueing the bolts and cap screws (kg)	
19.		Galvanisation	
	a)	Method of galvanisation	
		i) Spring Washers	
		ii) any other spare parts	
	b)	Quality of Zinc and standard to which conforming	
20.		Short circuit Rating (kA)	
21.		Equivalent compressive and tensile strength (kg)	
22		Electrical resistance of elastomer (ohms)	
23		Electrical resistance between jumper sub-conductors across the assembly (ohm)	
24		Maximum conductor temperature which can be with stood by the elastomer (deg. C.)	
25		Certification Mark (ISI/BS/or any other International Standard)	
26		Packing details	

Signature of Bidder
With Seal and Date